

Modeling Vinasse Production Through System Dynamics: A Circular Economy Approach for Biorefinery Scenario Simulation

Abstract

Global ethanol production generates massive quantities of vinasse, a highly polluting effluent averaging 10–15 liters per liter of ethanol produced, creating critical waste management challenges due to its high organic load and acidic pH. Existing techno-economic assessments rely on static models that ignore dynamic feedback between waste generation and economic reinvestment, leaving a methodological gap for long-term strategic planning. This study develops a structurally validated system dynamics model coupling agricultural production, ethanol manufacturing, and multi-pathway vinasse valorization through biogas, butyric acid, and water recovery. Structural validity is established through unit verification, parameter checking, and three extreme condition tests prior to behavior validation. The model is validated against 23 years of Brazilian historical data (2000–2023) with in-sample Mean Absolute Percentage Errors of 7.98% for ethanol and 4.82% for crop production, and out-of-sample errors of 8.48% and 6.30% respectively, all below the 10% high-accuracy threshold. Five scenarios were simulated over a 30-year horizon with long-term margins stabilizing between 8.91% and 17.76% and cumulative profits ranging from 22,395 to 61,638 MMUSD. Biogas achieves the highest profitability (17.76%, 61,638 MMUSD), followed by water recovery (12.34%, 31,891 MMUSD), butyric acid (8.91%, 30,931 MMUSD), and the diversified portfolio (10.57%, 22,395 MMUSD). Formal optimization via OptQuest converged to a corner solution allocating all vinasse to biogas, confirming that targeted single-pathway planning outperforms equal-weight diversification by 175% in cumulative profit due to resource allocation trade-offs in supply-constrained biorefineries. The biogas pathway advances SDG 7 by displacing fossil fuels, while water recovery advances SDG 6 by reducing pollutant discharge and recovering reusable irrigation water. These findings provide an evidence-based decision support framework confirming that circular economy approaches substantially improve the sustainability of the sugarcane-ethanol sector.

Keywords: Vinasse, System Dynamics, Circular Economy, Biorefinery, Simulation, Biogas

1 Introduction

Global progress toward sustainable development remains critically uneven: billions of people still lack access to safe drinking water and sanitation, while an escalating energy crisis continues to obstruct the transition toward affordable and clean energy for all, placing the Sustainable Development Goals (SDGs), particularly SDG 6 (Clean Water and Sanitation) and SDG 7 (Affordable and Clean Energy), among the most urgent and interconnected challenges of the coming decade (Bisht, Shah, and Rana (2023); United Nations Department of Economic and Social Affairs (2025)). To address these challenges, the bioethanol industry emerges as a pivotal solution by offering a renewable alternative to fossil fuels that significantly reduces greenhouse gas emissions and bolsters global energy security (Gurgel et al. (2024)). Beyond its role in the energy sector, the industry fosters a circular economy by repurposing bio-industrial residues into high-performance materials designed to remove toxic pollutants from wastewater, thereby ensuring the protection of water resources (Eltaher et al. (2026)).

Global bioethanol production reached approximately 61.0 billion liters in 2024, with Brazil contributing 20% as the second largest producer after the United States (Statista (2025)). *Sugarcane-based ethanol*, defined as ethanol produced through the fermentation of sugarcane juice and molasses, dominates the Brazilian biofuel sector, generating 33.3 billion liters in 2024 and supporting rural employment while reducing fossil fuel dependence (Statista (2025)). However, this large-scale production creates a critical waste management challenge: ethanol distillation generates a high-volume liquid by-product known as *vinasse*, with approximately 10 to 15 liters produced per liter of ethanol, resulting in 300 to 500 billion liters of this effluent annually in Brazil alone, raising critical environmental concerns including groundwater contamination, soil salinization, and greenhouse gas emissions (Borges et al. (2024); Reis and Hu (2017)).

Vinasse is characterized by a high organic load and an acidic pH whose nitrogen and mineral content support crop growth, making fertigation a common disposal practice (Reis and Hu (2017)). Given the potential to recover valuable components from this effluent, circular economy frameworks have been explored in the literature, describing closed-loop systems (Latiza and Rubi (2024)) designed to minimize waste by reintegrating residues into the production cycle through water reuse, nutrient recovery, and energy production. Experimental studies have identified three principal valorization pathways: organic acid recovery through low-cost treatment processes (Vidal, Poblete, Godoy, Olivares, and Salazar-González (2025)), high-quality water recovery for recirculation in industrial processes (Oliveira, Costa, and Costa (2025)), and biogas production through anaerobic digestion as a renewable energy source (Sica, Marabesi, Seleglim, Das, and Baptista (2023)). Nevertheless, the implementation of these circular economy strategies faces challenges related to economic viability, technical constraints, and social impacts.

Despite extensive research on individual valorization technologies, existing *Techno-Economic Assessments* (TEA) exhibit three critical methodological weaknesses (Bartling et al. (2021); Melin et al. (2022); Pandit et al. (2023)). Studies often evaluate pathways in isolation using static discounted cash flow analysis, ignoring dynamic feedback effects such as how biogas profitability influences reinvestment in

production capacity (Melin et al. (2022)). Analyses typically assume constant prices and stable demand, failing to account for market saturation effects as by-product supply increases (Pandit et al. (2023)). Comparative frameworks that simultaneously evaluate multiple valorization strategies under identical assumptions are also absent, preventing stakeholders from systematically comparing pathways or designing portfolios that balance risk and return (Bartling et al. (2021)). These limitations are particularly problematic for long-term strategic planning in the sugarcane-ethanol sector, where investment decisions must account for evolving markets, policy changes, and technology learning curves.

System dynamics modeling (SDM) offers a methodological framework specifically suited to address these issues (Sterman (2000)). Unlike static optimization models, SDM explicitly represents feedback loops, time delays, and non-linear relationships that govern biorefinery system evolution. In circular economy contexts, SDM has successfully modeled lithium-ion battery recycling (Pratap, Mohan, Amit, and Venugopal (2024)), municipal waste management (Vega and Bautista-Rodriguez (2024)), and sustainable logistics (Jayarathna, Agdas, and Dawes (2023)). In the bioethanol sector, SDM has been applied to assess the impact of federal policies on production expansion in the United States (Newes et al. (2021)), to project bioethanol blending scenarios in Australia (Taheri, Jahani, and Pishvaei (2024)), and to analyze ethanol blending programs integrating first- and second-generation production pathways (Kumar, Padhy, Das, and Narayana (2024)). However, current bioethanol SDM applications focus almost exclusively on the energy sector, omitting mechanisms needed to represent diverse product portfolios and the specific interactions of process industries (Latiza and Rubi (2024)). A critical gap remains in the development of models that incorporate comprehensive feedback loops in which the economic and technical feasibility of by-product valorization actively regulates system behavior and investment decisions over time (Roy and Tu (2022)).

This study addresses these methodological and empirical gaps by developing an integrated SDM of the sugarcane-ethanol vinasse system, coupling agricultural production, ethanol manufacturing, and multi-pathway vinasse valorization, and making four primary contributions:

- **An SDM-based By-product Production Module (BPM) dynamically coupled to the ethanol production system.** The BPM operationalizes the circular valorization of vinasse through biogas, butyric acid, and water recovery pathways, receiving a time-varying vinasse supply endogenous to the ethanol production dynamics. Although the coupling is unidirectional — ethanol production drives vinasse generation, which feeds byproduct revenues, without reverse feedback into upstream planting decisions — this architecture enables scenario analysis that captures the co-evolution of production scale and valorization economics over multi-decade horizons, a capability absent in static TEA frameworks (Latiza and Rubi (2024); Roy and Tu (2022)). The full mathematical formulation of the BPM and all model modules is provided in Appendix A.
- **A structurally validated SDM with 23-year empirical behavior validation using Brazilian sugarcane-ethanol historical data.** Prior to behavior validation, structural validity is established through systematic unit checking,

parameter verification, and three extreme condition tests, following the methodological sequence recommended by [Stermann \(2000\)](#). The model is then validated against 23 years of Brazilian historical data (2000–2023), achieving in-sample MAPEs of 7.98% for ethanol production and 4.82% for crop production, and out-of-sample MAPEs of 8.48% and 6.30% respectively, all below the 10% high-accuracy threshold for complex socio-technical SDMs. Calibration procedures and accuracy metrics are detailed in Appendix [B](#).

- **A five-scenario simulation revealing that targeted biogas investment outperforms equal-weight diversification.** A set of comparative simulations across five valorization scenarios — Biogas, Butyric Acid, Water Recovery, an integrated Diversification pathway, and a formally optimized portfolio allocation — over a 30-year horizon reveals that biogas achieves the highest long-run margin (17.76%) and cumulative profit (61,638 MMUSD), while equal-weight diversification stabilizes 7.2 pp below the biogas ceiling. Formal optimization via OptQuest converges to a corner solution allocating all vinasse to biogas, confirming that targeted single-pathway planning outperforms equal-weight diversification due to resource allocation trade-offs in supply-constrained biorefineries ([Jayarathna et al. \(2023\)](#); [Vega and Bautista-Rodriguez \(2024\)](#)).
- **An open-access AnyLogic model for replication and extension of all reported scenarios.** The full implementation of the system dynamics model, including all modules, parameterizations, and scenario configurations, is publicly available at <https://doi.org/10.5281/zenodo.18983277>. This open-access repository enables independent replication of all reported results, facilitates sensitivity testing under alternative parameter assumptions, and provides a ready-to-use platform for researchers and practitioners seeking to extend the model to other biorefinery contexts or regional settings.

The study is organized as follows. Section [2](#) reviews relevant literature on vinasse valorization technologies, techno-economic assessments of biorefinery pathways, and SDM applications in circular economy contexts. Section [3](#) presents the model development methodology, including causal loop diagrams, stock-flow formulation, structural validity tests, and behavior validation procedures. Section [4](#) reports simulation results for each scenario alongside their interpretation, comparing financial performance trajectories and sensitivity analyses against existing literature and addressing strategic implications and model limitations. Section [5](#) concludes with key insights, practical recommendations, and directions for future research.

2 Literature Review

This literature review examines three interconnected research domains that inform the development of an integrated SDM model for vinasse valorization: vinasse characteristics and valorization technologies, techno-economic assessment methodologies for biorefinery pathways, and SDM applications in circular economy and biorefinery contexts.

2.1 Vinasse Characteristics and Valorization Technologies

Vinasse, the primary residual stream from ethanol distillation, presents both environmental challenges and opportunities for resource recovery. [Reis and Hu \(2017\)](#) characterize vinasse by high organic loads (COD 50 to 150 g/L), acidic pH (3.5 to 5.0), elevated temperatures (85 to 90°C), and significant concentrations of potassium, nitrogen, and phosphorus. While these nutrient characteristics have historically supported fertigation in sugarcane cultivation, [Borges et al. \(2024\)](#) demonstrate that intensive application leads to soil salinization, groundwater contamination, and greenhouse gas emissions from uncontrolled organic matter decomposition, catalyzing research into valorization pathways that capture value while mitigating environmental risks.

Biogas production through anaerobic digestion represents the most technologically mature pathway. [Fu, Xu, Dai, Yuan, and Guo \(2017\)](#) report methane yields of 0.37 L CH₄/g COD removed using upflow anaerobic sludge blanket (UASB) reactors, while [Ferraz, Etchebehere, Perecin, Teixeira, and Woods \(2022\)](#) estimate a yield of 12.115 Nm³ of biogas per cubic meter of vinasse processed, equivalent to 21.8 MJ of energy content. [Sica et al. \(2023\)](#) confirm positive net present values for biogas valorization when integrated with existing mill operations, particularly when displacing fossil fuels for steam generation or upgrading to biomethane.

Organic acid production represents an emerging high-value pathway for the ethanol industry. [De Souza Moraes, Santos, Delforno, Fuess, and Da Silva \(2018\)](#) achieved butyric acid concentrations of 10.6 g/L through dark fermentation using enriched microbial consortia, while [Fuess, Santos, Delforno, De Souza Moraes, and Da Silva \(2020\)](#) report superior energy efficiency compared to biogas when considering the economic value of chemical versus energy products. The global butyric acid market, valued at approximately \$231 million in 2023, is projected to grow at 6.2% annually through 2030, driven by demand in polymer, pharmaceutical, and animal nutrition applications ([Suazo, Tapia, Aroca, and Quintero \(2023\)](#)).

Water recovery technologies address freshwater scarcity in sugarcane-growing regions. [Peiter, Hankins, and Pires \(2019\)](#) demonstrated that up to 70% of vinasse volume can be recovered as distilled water suitable for industrial reuse using concentration technologies, while [Oliveira et al. \(2025\)](#) achieved exergy efficiencies of 68% with an evaporation-based system optimized via genetic algorithms, reducing freshwater consumption by 1.2 m³ per ton of processed sugarcane. Economic viability remains sensitive to local water prices and energy costs.

Despite these technological advances, existing literature examines valorization pathways in isolation, without systematic comparison of their relative economic performance, scalability constraints, or synergistic potential within biorefinery portfolios.

2.2 Techno-Economic Assessment of Biorefinery Pathways

TEA has emerged as the dominant methodology for evaluating biorefinery valorization strategies, yet current approaches exhibit three critical methodological limitations ([Bartling et al. \(2021\)](#); [Melin et al. \(2022\)](#); [Pandit et al. \(2023\)](#)). First, static discounted cash flow models cannot capture dynamic feedback effects between production scale, market development, and technology learning. [Melin et al. \(2022\)](#) evaluated

hybrid biomass-electricity ethanol production using fixed parameters and constant prices, ignoring how increased supply affects market prices through supply-demand dynamics — a limitation particularly consequential for emerging products such as bio-based butyric acid. Second, TEA studies assume stable market conditions throughout project lifecycles of 20 to 30 years, overlooking correlated parameter evolution such as how renewable energy policy changes simultaneously affect biogas prices, carbon credit values, and competing fossil fuel costs, as illustrated by Pandit et al. (2023). Third, pathways are evaluated independently rather than comparatively: Bartling et al. (2021) conducted a comprehensive TEA of reductive catalytic fractionation biorefineries focused on a single technology configuration, leaving stakeholders without the frameworks needed to rank pathways or design portfolios that balance risk and return.

Recent attempts to address these limitations through Monte Carlo simulation or scenario analysis still operate within static equilibrium frameworks. What remains absent is a methodology that endogenously represents how valorization pathway performance feeds back to influence upstream production decisions, technology adoption rates, and market evolution over multi-decade timeframes.

2.3 SDM in Circular Economy and Biorefinery Systems

SDM offers a complementary methodology designed to represent feedback-driven, time-dependent system behavior (Stermann (2000)). In circular economy contexts, SDM has modeled lithium-ion battery recycling, demonstrating that viability depends on achieving sufficient collection volumes to justify processing investments, with consumer participation responding to regulatory incentives with 2 to 5 year lags (Pratap et al. (2024)). SDM has also evaluated municipal solid waste management, identifying critical thresholds where policy ambition outpaces market absorption capacity (Vega and Bautista-Rodriguez (2024)), and assessed sustainable logistics, revealing that aggressive multi-pathway expansion without corresponding demand depth reduces system profitability (Jayarathna et al. (2023)).

Within the bioethanol sector, SDM applications have focused primarily on production scaling and policy impacts. The Biomass Scenario Model captured how policy-driven demand growth in the United States triggered capacity investments with 3 to 7 year construction delays (Newes et al. (2021)), while similar approaches have projected sustainable blending scenarios in Australia under food-energy security constraints (Taheri et al. (2024)) and analyzed coordinated investment requirements for ethanol blending programs in India (Kumar et al. (2024)). However, these models focus almost exclusively on fuel production, omitting by-product valorization pathways and their feedback effects on overall biorefinery economics. Roy and Tu (2022) identify this as a critical gap in a systematic review, noting the absence of multi-product stream models with explicit market development dynamics for emerging bio-based products.

The review identifies three gaps motivating the present study: the absence of endogenous feedback loops linking vinasse valorization economics to upstream production decisions, the scarcity of long-term model validation against historical data, and the lack of comparative scenario analysis evaluating multiple valorization pathways under identical assumptions. The present study addresses all three through the

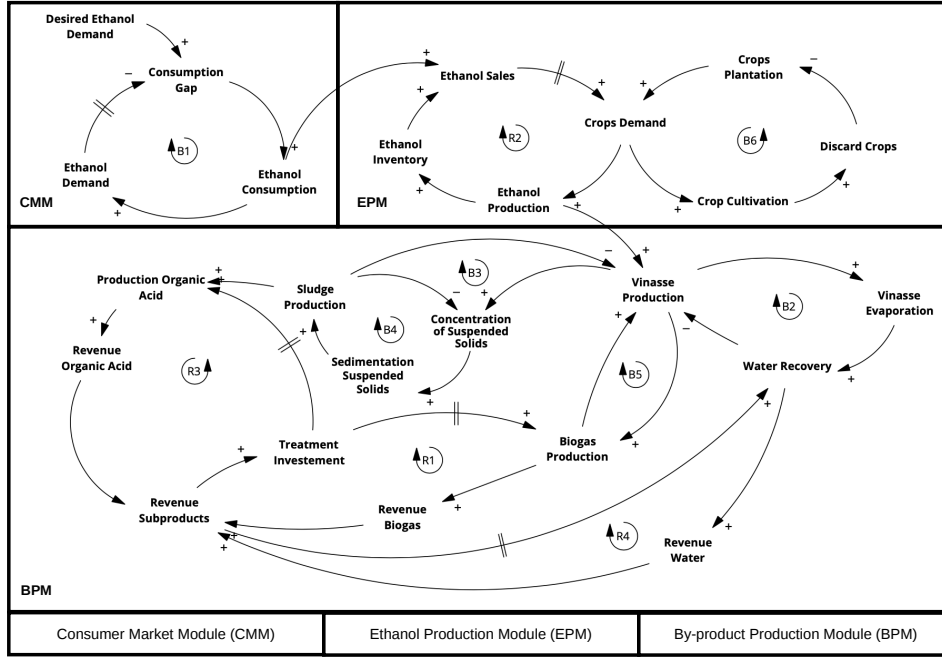


Fig. 1: Causal loop diagram (CLD) of the ethanol market and by-product production system.

By-product Production Module, a 23-year empirical validation, and a five-scenario comparative simulation framework.

3 Methodology

This study develops a system dynamics model (SDM) to evaluate the technical and economic feasibility of transitioning from linear to circular biorefinery operations, with a focus on vinsasse valorization. The model integrates four interconnected modules simulating the coupled dynamics of the ethanol market, crop production, ethanol and vinsasse generation, and byproduct market development, following the five-stage modeling process of [Sterman \(2000\)](#).

3.1 Formulating Dynamic Hypotheses: Causal Loop Diagrams

Causal loop diagrams (CLDs) ([Sterman \(2000\)](#)) represent the dynamic interactions between the ethanol market and byproduct production, organized around three interconnected modules: the Ethanol Production Module (EPM), the By-product Production Module (BPM), and the Consumer Market Module (CMM), as shown in Figure 1.

The CMM captures demand-side dynamics through a balancing loop (B1): the gap between desired ethanol demand and actual ethanol consumption drives a negative feedback adjustment in ethanol demand, which in turn governs consumption. As demand increases, consumption rises to close the gap, reducing the gap and dampening further demand growth. This self-correcting structure ensures that consumption converges toward desired demand rather than growing without bound, representing the stabilizing nature of consumer markets. Every arc in B1 carries a positive polarity except the link from consumption back to the gap, which is negative, producing the net balancing behavior.

Within the EPM, a reinforcing loop (R2) operates as follows: higher ethanol inventory enables greater sales, which generates crop demand, which drives ethanol production, which replenishes inventory. Every arc in R2 carries a positive polarity, producing self-reinforcing growth in market activity and agricultural expansion. A balancing loop (B6) imposes a natural limit on this growth through the crop lifecycle: plantation increases young crop stocks, which mature into harvest-ready crop, which is eventually discarded at a rate proportional to the crop stock. The discard flow reduces the plantation stock, closing B6 with a net negative feedback that prevents unbounded agricultural accumulation.

The BPM operationalizes the circular valorization of vinasse through three parallel pathways. Three reinforcing loops (R1, R3, R4) capture the internal economics of each valorization route: revenues from biogas, organic acid, and water recovery feed into a shared treatment investment stock, which expands processing capacity and further increases production across all three pathways, generating additional revenues. Each of R1, R3, and R4 is a positive feedback loop in which higher revenue drives investment, investment drives output, and output drives further revenue. Balancing loops (B2–B5) represent physical constraints within the BPM: vinasse evaporation reduces available storage (B2), suspended solids concentration and sedimentation govern sludge production (B3, B4), and the allocation of vinasse to biogas and water recovery pathways creates competing claims on the available effluent stock (B5), each imposing a negative feedback that limits the growth of the respective pathway.

The causal structure reflects a **unidirectional coupling** from the EPM to the BPM: ethanol production determines vinasse generation, which drives byproduct revenues, but byproduct revenues do not feed back into upstream agricultural or ethanol production decisions. This boundary is consistent with the single-facility modeling scope, in which valorization pathway economics are evaluated conditional on the production dynamics of the sugarcane-ethanol system rather than as active regulators of those dynamics. The model’s contribution is therefore the dynamic coupling of a time-varying vinasse supply — endogenous to the ethanol production system — with a multi-pathway economic valorization module, enabling scenario analysis that captures the co-evolution of production scale and valorization revenues over multi-decade horizons.

3.2 Model Formalization: Stock-and-Flow Diagrams

The CLDs were translated into quantitative Stock-and-Flow Diagrams (SFDs) (Stermann (2000)) organized into four interconnected modules. The CMM models demand-side population dynamics driving ethanol consumption using a lookup table interpolated from historical and projected Brazilian population data (2000–2030), replacing the exponential birth-death formulation with an empirically grounded trajectory that captures the observed deceleration in Brazilian population growth over the simulation horizon; the Ethanol Inventory Module handles production, inventory management, and market-clearing logic; the Crop Production Module models agricultural biomass stocks with feedback-controlled planting; and the BPM operationalizes the circular valorization of vinasse into byproducts, treating vinasse generation as an auxiliary variable computed from ethanol production in each period rather than as an accumulated stock, consistent with the continuous-flow processing assumption of industrial distillery operations. The full mathematical formulation of each module, including all stock-and-flow equations, units, and diagrams, is provided in Appendix A.

3.3 Model Testing: Structural Validity and Behavior Validation

Structural validity is established prior to behavior validation, following the methodological sequence recommended by Stermann (2000): structure verification, parameter verification, extreme condition tests, and behavior validity assessment.

3.3.1 Structure Verification

Model verification involved systematic unit checking across all equations, assigning explicit units to every variable and flow to confirm dimensional consistency. All variables and flows were verified to carry consistent units throughout the four modules; the full unit specification is provided in Appendix A.

3.3.2 Parameter Verification

All parameters have real-world counterparts documented in Table 4. Population dynamics are represented through a lookup table interpolated from historical Brazilian census data and demographic projections (2000–2030), capturing the observed transition from accelerating to decelerating growth without imposing an exponential structural form. Six agricultural parameters are reported in Table 4: four were estimated through calibration against historical crop production data (*AdjustTimeCrop*, *AdjustTimeYoungCrop*, *AverageDiscardTime*, *GrowthDelayTime*), and two are explicit modeling assumptions (*ProcessingTime* = 1 year and *SalesTime* = 1 year) reflecting the annual planning and sales cycle of the Brazilian sugarcane-ethanol sector (Newes et al. (2021); Taheri et al. (2024)); all values fall within agronomically and operationally plausible ranges derived from the literature (Table 4). Economic and technical parameters for the BPM are drawn directly from peer-reviewed sources without data-driven estimation.

3.3.3 Extreme Condition Tests

Three extreme condition tests were conducted to verify structural consistency under boundary scenarios (Sterman (2000); Vega and Bautista-Rodriguez (2024)), following the protocol of applying theoretically limiting parameter values to the whole model and verifying that the model produces structurally plausible responses. For each test, key variables are reported at selected time points ($t = 0, 5, 8, 10, 15, 20, 25, 30$) to document both instantaneous values and dynamic trajectories.

Test 1 — Zero Valorization ($ProportionBiogas = 0$, $ProportionOrganicAcid = 0$, $ProportionWater = 0$): all byproduct flows are forced to zero, leaving ethanol sales as the sole revenue source. Table 1 confirms that BiogasProduction, OrganicAcidProduction, and WaterRecovery remain identically zero throughout the 30-year horizon, while the agricultural-ethanol subsystem continues operating undisturbed. Net margin grows from 6.94% at Year 8 to 10.57% at Year 30, driven entirely by ethanol revenue, confirming that the BPM is fully decoupled from the upstream modules and that removing valorization does not destabilize core production dynamics.

Table 1: Test 1 — Zero Valorization: byproduct production, net profit margin, and crop stock across selected time periods. All byproduct proportions set to zero throughout the 30-year horizon. Crop in thousand tons.

Year	Biogas (Nm ³ /yr)	Org. Acid (kg/yr)	Water Rec. (m ³ /yr)	Net Margin (%)	Crop (th. ton)
2000	0	0	0	—	0.0
2005	0	0	0	1.65	457,199.8
2008	0	0	0	6.94	543,704.5
2010	0	0	0	8.44	574,093.4
2015	0	0	0	9.76	612,477.5
2020	0	0	0	10.19	632,688.3
2025	0	0	0	10.50	645,587.3
2030	0	0	0	10.57	654,673.0

Test 2 — Zero Population Growth (population held constant at 174,018,282 persons; all pathways active at equal weights 1/3): Table 2 shows that byproduct flows stabilize progressively — Biogas reaches 1,164,647 Nm³/yr and OrganicAcid 1,019,006 kg/yr at Year 30, with marginal increments between Year 25 and Year 30 of 461.0 Nm³/yr and 403.3 kg/yr respectively, confirming convergence toward a dynamic equilibrium. Crop stock growth decelerates from +1,045.3 to +228.5 thousand tons between the Y20–Y25 and Y25–Y30 intervals, approaching steady state at 577,373 thousand tons. Net margin stabilizes between −10.14% and −10.17% from Year 25 onward, confirming that the growth dynamics observed in the main scenarios originate exclusively from the population-driven demand trajectory encoded in the lookup table.

Table 2: Test 2 — Zero Population Growth: byproduct production, net profit margin, and crop stock across selected time periods. Population held constant at 174,018,282 persons; all three valorization pathways active at equal weights (1/3 each). Crop in thousand tons.

Year	Biogas (Nm ³ /yr)	Org. Acid (kg/yr)	Water Rec. (m ³ /yr)	Net Margin (%)	Crop (th. ton)
2000	0.0	0.0	0.0	—	0.0
2005	906,507	793,147	52,378	−7.98	449,400.5
2008	1,061,051	928,365	61,307	−4.23	526,015.5
2010	1,108,314	969,717	64,038	−6.54	549,446.0
2015	1,152,432	1,008,319	66,587	−9.39	571,317.7
2020	1,162,077	1,016,758	67,144	−10.00	576,099.2
2025	1,164,186	1,018,602	67,266	−10.14	577,144.5
2030	1,164,647	1,019,006	67,293	−10.17	577,373.0

Test 3 — Zero Ethanol Yield ($ethanolYield = 0$): domestic ethanol production and all byproduct flows collapse to zero. Table 3 verifies that Imports = Shortage in every period, confirming correct functioning of the import mechanism. The Ethanol stock grows from zero because imports accumulate in inventory before being sold; Sales = Ethanol/ $SalesTime$ in each period, consistent with the market-clearing formulation in Appendix A. Shortage grows steadily from 14,346.1 to 16,290.0 thousand m³/year between Year 8 and Year 30, tracking the population-driven increase in DemandVolume/ $SalesTime$. Revenue is negative relative to cost throughout the horizon, confirming both the causal integrity of the EPM-to-BPM chain and the correct functioning of the import mechanism as a supply backstop under complete domestic production failure.

All three tests produced structurally consistent results, satisfying the structural validity requirements recommended by [Sterman \(2000\)](#) before proceeding to behavior validation.

3.3.4 Behavior Validation

The model was parameterized using data from the Brazilian sugarcane-ethanol sector for the period 2000 to 2023, supplemented with values from the literature and technical reports. Key parameters are summarized in Table 4. Per capita ethanol consumption was estimated as the ratio of national ethanol production to total population for each year. The first eight years (2000–2007) were used as a warm-up period to initialize the model stocks before the calibration window. Following the warm-up period, a split-sample validation approach was used: the model was calibrated on data from 2008 to 2018 and evaluated on an independent out-of-sample period from 2019 to 2023. Accuracy was assessed using the Mean Absolute Percentage Error (MAPE) ([Akhavan and Gonçalves \(2021\)](#); [De Myttenaere, Golden, Grand, and Rossi \(2016\)](#); [Yuniaristanto,](#)

Table 3: Test 3 — Zero Ethanol Yield: ethanol stock, net profit margin, demand volume, imports, and ethanol production across selected time periods. *ethanolYield* = 0; domestic ethanol production and all byproduct flows collapse to zero throughout the horizon. Ethanol stock, DemandVolume, Imports in thousand m³; Ethanol production in thousand m³/yr.

Year	Eth. Stock (th. m ³)	Net Margin (%)	DemandVolume (th. m ³)	Imports (th. m ³)	Eth. Production (th. m ³ /yr)
2000	0.0	—	26,102.7	26,102.7	0
2005	13,775.3	−15.87	27,703.2	13,927.9	0
2008	14,209.0	−14.17	28,555.1	14,346.1	0
2010	14,466.3	−13.50	29,055.3	14,589.0	0
2015	15,065.9	−12.74	30,251.3	15,185.4	0
2020	15,603.6	−12.14	31,299.1	15,695.5	0
2025	15,930.2	−11.69	31,921.9	15,991.7	0
2030	16,230.0	−11.51	32,520.0	16,290.0	0

(Sutopo, Hisjam, and Wicaksono (2024)), defined in Equation B26 (Appendix B). In the context of SDM validation, MAPE below 10% is considered highly accurate, 10% to 20% indicates acceptable accuracy for complex socio-technical systems, and above 20% requires recalibration (Sterman (2000)). Validation results are reported in Section 4.1.

3.4 Scenario Design for Comparative Analysis

Five distinct scenarios were designed to assess the techno-economic impact of vinasse valorization, spanning from single-pathway valorization to integrated portfolios and a formally optimized allocation, and were implemented in AnyLogic 8.9.6 PLE. The simulation horizon was set to 30 years for three reasons: consistency with investment planning cycles of 20 to 30 years for capital-intensive valorization infrastructure (Bartling et al. (2021); Pandit et al. (2023)); alignment with RenovaBio program of Brazil, which projects decarbonization targets through 2050 (Sica et al. (2023)); and the requirement to observe stabilization of feedback-driven dynamics, given biological maturation delays of approximately 1.095 years and average discard times of approximately 5.319 years in the crop module (Table 4), requiring multi-decade simulation windows to reach steady state (Sterman (2000)).

3.4.1 Scenario 1: Biogas Valorization

Scenario 1 integrates anaerobic digestion to convert vinasse into biogas using upflow anaerobic sludge blanket (UASB) reactors (Fu et al. (2017)), establishing an additional revenue stream from energy recovery while reducing disposal costs (Nair and Taherzadeh (2016)). The scenario assumes a technologically feasible proportion

Table 4: Technical, economic, and demographic parameters of the sugarcane-ethanol biorefinery SDM. Calibrated parameters were estimated from historical data (2008–2018).

Parameter	Unit	Value	Reference
Technical parameters			
Ethanol usage	m ³ /person	0.15	Rate between ethanol production and population (2000–2023) UNICAdata (2024)
Ethanol yield	m ³ / ton	0.05	Rate between ethanol production and crop production (2000–2023) UNICAdata (2024)
Adjust time crop	years	3.590	Model calibration
Adjust time young crop	years	1.590	Model calibration
Average discard time	years	5.319	Model calibration
Growth delay time	years	1.095	Model calibration
Ethanol to crop ratio	ton/m ³	13.1	Shelar et al. (2023)
Butyric acid factor	g/L vinasse	10.6	De Souza Moraes et al. (2018)
Water recovery factor	dimensionless	0.7	Peiter et al. (2019)
Biogas factor	Nm ³ /m ³ vinasse	12.115	Ferraz et al. (2022)
Vinasse yield	m ³ /m ³ ethanol	10	Reis and Hu (2017)
Economic parameters			
Ethanol production cost	USD/m ³	423	Jonker et al. (2015)
Sugarcane planting cost	USD/ton	15.2	De Amorim and Patino (2022)
Ethanol import cost	USD/thousand m ³	414,680	Trading Economics (2026)
Import tariff	dimensionless	0.18	Barros (2025)
Ethanol price	USD/L	0.5455	CEPEA (2025)
Biogas price	USD/Nm ³	0.5	IRENA (2017)
Butyric acid price	USD/kg	2.89	Suazo et al. (2023)
Butyric acid production cost	USD/kg	2.75	Fuess et al. (2020)
Biogas production cost	USD/Nm ³	0.075	De Oliveira Carneiro et al. (2022)
Water cost treatment (Evaporator)	USD/m ³	1.03	Peiter et al. (2019)
Water cost	USD/m ³	2.81	Peiter et al. (2019)
Demographic parameters			
Population (2000–2025)	persons	Lookup table	Expansión/Datosmacro (2024)
Population (2026–2030)	persons	Lookup table	IMF (2025)

of vinasse is diverted to anaerobic digestion reactors operating under oxygen-free conditions.

3.4.2 Scenario 2: Butyric Acid Production

Scenario 2 evaluates butyric acid production from vinasse through dark fermentation, targeting chemical markets for bioplastics, animal feed additives, and pharmaceutical intermediates. Enriched microbial consortia metabolize vinasse organic compounds under controlled anaerobic conditions, yielding concentrations up to 10.6 g/L ([De Souza Moraes et al. \(2018\)](#)), with demand driven by bio-replacement needs in these sectors ([Fuess et al. \(2020\)](#)).

3.4.3 Scenario 3: Water Recovery

Scenario 3 models the integration of concentration technologies to recover reusable water from vinasse for crop irrigation, representing a closed-loop water management

strategy aligned with circular economy principles (Latiza and Rubi (2024)). The scenario is governed by a water recovery factor of 0.7 (Peiter et al. (2019)), with economic viability dependent on local water prices and energy costs.

3.4.4 Scenario 4: Diversification Scenario

The Diversification scenario evaluates an equal-weight portfolio in which available vinasse is partitioned equally (33.3% each) into biogas production, butyric acid fermentation, and water recovery. This symmetric allocation serves as a baseline for examining system behavior under balanced resource distribution, providing a reference point against which the formally optimized allocation of Scenario 5 can be compared.

3.4.5 Scenario 5: Optimized Portfolio Allocation

To formally identify the profit-maximizing allocation of vinasse, an optimization experiment was conducted using the OptQuest solver built into AnyLogic. The decision variables are *ProportionBiogas*, *ProportionOrganicAcid*, and *ProportionWater*, each bounded between 0 and 1, subject to the constraint that their sum does not exceed unity (Equation 1), allowing corner solutions and configurations where a fraction of vinasse remains unallocated.

$$\text{ProportionBiogas} + \text{ProportionOrganicAcid} + \text{ProportionWater} \leq 1 \quad (1)$$

The objective function maximizes the cumulative net profit Π over the 30-year horizon T (Equation 2), where R_{ethanol} , R_{biogas} , R_{acid} , and R_{water} represent revenues from each stream and C_{total} aggregates all operational costs.

$$\max \quad \Pi = \int_0^T (R_{\text{ethanol}} + R_{\text{biogas}} + R_{\text{acid}} + R_{\text{water}} - C_{\text{total}}) dt \quad (2)$$

OptQuest iteratively evaluates candidate allocation vectors over 500 iterations, updating the search direction based on the profit response surface until convergence.

3.5 Sensitivity Analysis

Sensitivity analysis evaluates model robustness across three variable categories, following methodologies established in circular economy SDM studies. The analysis examines price, cost, and factor conversion parameters at variation rates of $\pm 5\%$, $\pm 25\%$, $\pm 50\%$, and $\pm 100\%$.

For financial variables (prices and costs), these variation rates are applied to parameters such as biogas price, butyric acid cost, and water treatment costs. The $\pm 5\%$ level captures marginal short-term market fluctuations, $\pm 25\%$ represents normal commodity price volatility consistent with biogas and butyric acid markets, $\pm 50\%$ represents moderate stress scenarios such as policy reversals or supply chain disruptions, and $\pm 100\%$ represents extreme boundary conditions (Jayarathna et al. (2023)). This four-level structure follows sensitivity reporting conventions in circular economy SDM studies (Pratap et al. (2024); Vega and Bautista-Rodriguez (2024)), providing sufficient resolution to characterize the risk profile of each pathway. The same variation rates are applied to conversion factors and technical efficiency variables such as

the biogas factor ($12.115 \text{ Nm}^3/\text{m}^3$), organic acid factor (10.6 g/L) and water recovery factor (0.7), ensuring a consistent analytical framework across both financial and technical parameters.

4 Results

The simulation model was executed across all five scenarios over a 30-year horizon. All monetary values are in millions of United States dollars (MMUSD). The first eight years (2000–2007) constitute a warm-up period during which stocks initialize; the analysis of profitability therefore begins at Year 8 (2008), coinciding with the start of the calibration window. The BPM enables dynamic profitability trajectories to be observed rather than inferred from a single point-in-time calculation (Fuess et al., 2020; Peiter et al., 2019; Sica et al., 2023), coupling vinasse generation with parallel valorization pathways through a unidirectional causal structure in which the time-varying vinasse supply — endogenous to the ethanol production system — drives byproduct revenues over multi-decade horizons.

4.1 Model Behavior Validation

Figure 2 compares simulated and observed trajectories for ethanol and crop production across the calibration and out-of-sample phases. During calibration (2008–2018), in-sample MAPEs were 7.98% for ethanol production and 4.82% for crop production, both below the 10% high-accuracy threshold (Sterman (2000)). The elevated ethanol MAPE is largely attributable to 2011–2012, when the Brazilian sugarcane-ethanol industry experienced a documented structural crisis driven by adverse weather conditions, suppressed pump prices under government fuel policy, and accumulated sectoral debt, producing an abrupt production decline that a deterministic SDM cannot replicate without an exogenous shock variable; excluding these two years, the in-sample ethanol MAPE falls to approximately 5.2%. In the out-of-sample phase (2019–2023), MAPEs for crop production (6.30%) and ethanol production (8.48%) remained below the 10% high-accuracy threshold, indicating an absence of overfitting and confirming model robustness for long-term scenario analysis.

4.2 Scenario Comparison

Figure 3 presents the cost, revenue, and net profit margin trajectories for the four valorization scenarios. All pathways enter the post-warmup period at Year 8 already operating with positive margins, confirming that the initialization dynamics do not bias the profitability analysis. The recalibrated model, validated against 23 years of Brazilian historical data (Section 4.1), yields in-sample MAPE values of 7.98% for ethanol production and 4.82% for crop production, with out-of-sample errors of 8.48% and 6.30% respectively, ensuring that 30-year projections rest on a structurally validated foundation. The comparative trajectories reveal a consistent profitability ranking that persists throughout the 30-year horizon: biogas dominates across all time periods, followed by water recovery, diversification, and butyric acid.

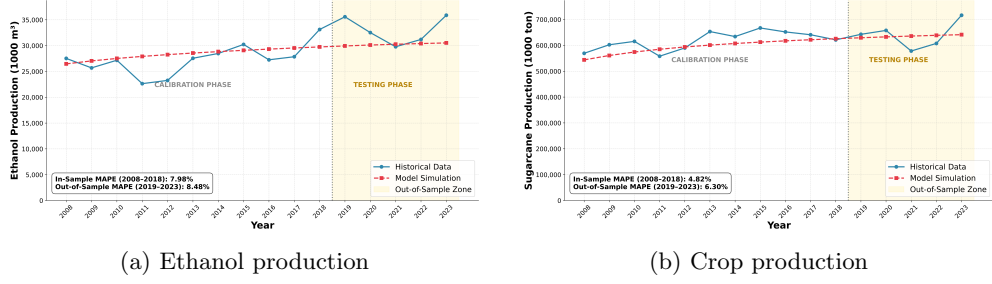


Fig. 2: Model validation results comparing observed and simulated time series under a split-sample approach (Calibration: 2008–2018; Out-of-Sample Testing: 2019–2023) for: (a) ethanol production (in-sample MAPE: 7.98%; out-of-sample MAPE: 8.48%) and (b) crop production (in-sample MAPE: 4.82%; out-of-sample MAPE: 6.30%).

Table 5 summarizes the key financial indicators for all four scenarios across the post-warmup horizon. At Year 8, net profit margins span a range of 7.6 pp across scenarios, from 6.80% (butyric acid) to 14.36% (biogas), with water recovery (8.80%) and diversification (6.94%) occupying intermediate positions. This early divergence widens over time: by Year 15, the gap between biogas (17.06%) and butyric acid (8.37%) reaches 8.7 pp, and by Year 30 it stabilizes at 8.8 pp (17.76% versus 8.92%). The ranking remains invariant throughout the horizon, indicating that the profitability hierarchy is determined by structural economics — principally the price-cost ratio of each pathway — rather than by transient initialization effects.

Table 5: Comparative financial indicators across valorization scenarios. Net profit margins at key years and cumulative profit over the full 30-year simulation horizon (Y0–Y30). SDG alignment reflects the primary sustainable development contribution of each pathway.

Scenario	Margin Y8	Margin Y15	Margin Y30	Cumul. Profit	SDG Alignment
Biogas	14.36%	17.06%	17.76%	55,706 MMUSD	SDG 7
Organic Acid	6.80%	8.37%	8.92%	32,369 MMUSD	—
Water Recovery	8.80%	11.57%	12.34%	25,372 MMUSD	SDG 6
Diversification	6.94%	9.76%	10.57%	15,688 MMUSD	SDG 6+7

The revenue trajectories are structurally informative. Revenue growth between Years 8 and 30 is 18% for biogas, water recovery, and diversification, and 19% for butyric acid, reflecting the common driver of population-linked ethanol demand growth that scales byproduct production proportionally across all pathways. The differentiation in long-run margins therefore originates entirely from cost trajectories: biogas

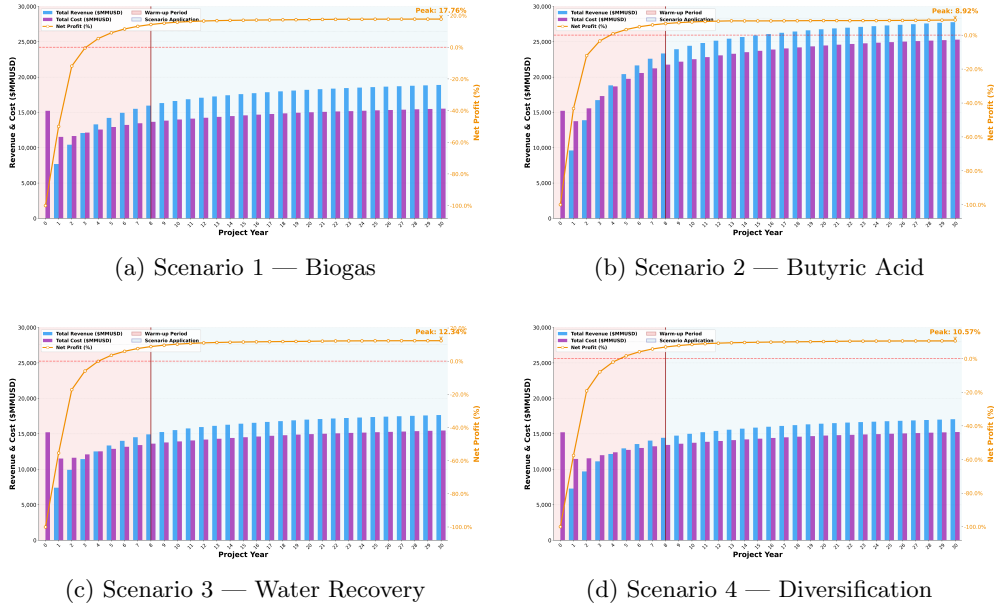


Fig. 3: Total costs, revenue, and net profit margin for each valorization scenario over the 30-year simulation horizon: (a) Biogas, (b) Butyric Acid, (c) Water Recovery, and (d) Diversification. All monetary values in million USD (MMUSD). The first eight years correspond to the model warm-up period.

and water recovery exhibit cost growth of 14% over the same period, while butyric acid reaches 16% and diversification 14%. The steeper cost trajectory of butyric acid reflects the processing-intensity of dark fermentation — microbial consortium management, controlled anaerobic conditions, and downstream purification — which scales more steeply with volume than the UASB reactor infrastructure underlying biogas or the evaporation-based system underlying water recovery.

The biogas pathway achieves the highest long-run margin (17.76% at Year 30) due to its exceptionally wide price-cost spread: at USD 0.074/Nm³ production cost against a USD 0.50/Nm³ sale price, biogas captures a gross unit margin ratio of approximately 6.8:1 that no other pathway approaches (Ferraz et al., 2022; Fu et al., 2017). Water recovery stabilizes at 12.34% by Year 30, supported by the USD 2.81/m³ water price generating a unit margin of USD 1.78/m³ recovered — substantially wider than butyric acid — and by the unique feature that the pathway simultaneously generates revenue and reduces a variable input cost in the ethanol production system through water recirculation for crop irrigation (Oliveira et al., 2025; Peiter et al., 2019). Butyric acid, with a unit margin of only USD 0.14/kg (a price-cost ratio of 1.05:1), stabilizes at 8.92%, confirming that high revenue growth does not translate into superior profitability when cost trajectories are steep and margins are thin (Fuess et al., 2020).

The diversification scenario (10.57% at Year 30) underperforms not only biogas but also water recovery, despite activating all three reinforcing loops in the BPM simultaneously. The equal-weight allocation of 33.3% per pathway mechanically caps each route’s contribution: biogas provides approximately 5.9 pp of the diversified margin, water recovery approximately 4.1 pp, and butyric acid approximately 3.0 pp, leaving a portfolio 7.2 pp below the biogas ceiling. This outcome reflects a structural feature of supply-constrained biorefineries that distinguishes them from financial portfolios: the three pathways are not uncorrelated assets but compete for the same upstream vinasse supply, meaning that the supply constraint binds jointly rather than independently (Bartling et al., 2021). As identified by Jayarathna et al. (2023) in sustainable logistics SDMs and by Vega and Bautista-Rodriguez (2024) in municipal waste management, aggressive multi-pathway expansion without corresponding demand depth reduces system profitability relative to targeted single-pathway investment. The diversified portfolio does, however, offer a lower risk profile: at $\pm 100\%$ price variation, its 20.30 pp sensitivity range is less than half the 54.83 pp range of butyric acid alone, making diversification a rational choice for operators facing extreme price uncertainty even at the cost of long-run profitability.

Beyond financial performance, two pathways carry direct sustainable development implications (Table 5). Biogas advances SDG 7 (Affordable and Clean Energy) by displacing fossil fuels for steam generation at the mill level, reducing greenhouse gas emissions without relying on external energy markets (Sica et al., 2023). Water recovery advances SDG 6 (Clean Water and Sanitation) by simultaneously reducing pollutant discharge from vinasse disposal and recovering reusable irrigation water that lowers freshwater extraction in sugarcane-growing regions (Latiza & Rubi, 2024; Peiter et al., 2019). Butyric acid does not align directly with either goal under current market conditions, reinforcing its position as a commercially contingent rather than strategically priority pathway. The diversification scenario partially addresses both SDGs through its combined allocation, but at reduced scale for each individual pathway relative to the targeted single-pathway alternatives.

4.3 Scenario 5: Optimized Portfolio Allocation

OptQuest converged after 500 iterations to a corner solution: $ProportionBiogas = 1$, $ProportionOrganicAcid = 0$, $ProportionWater = 0$. Since this replicates Scenario 1 exactly, the optimizer returns the same cumulative profit as the biogas scenario: 70,272 MMUSD over the 30-year horizon, which is 93% higher than the diversification scenario (36,487 MMUSD) and 57% higher than water recovery (44,663 MMUSD). The key finding is that no reallocation of vinasse proportions under current market parameters can close this gap, confirming that the underperformance of equal-weight allocation is structural rather than a consequence of suboptimal parameter choices.

A parametric threshold analysis shows that butyric acid prices would need to reach USD 3.85/kg (+33% over the USD 2.89/kg base case) to match the 17.76% long-run biogas margin, a level exceeding current market projections through 2030 (Suazo et al., 2023). This threshold is not a marginal tipping point: it requires a simultaneous price increase of 33% above the base case and a biogas price decline below USD 0.35/Nm³ (−30% from the base case of USD 0.50/Nm³) for any mixed allocation to outperform

full biogas dedication. The two-dimensional sensitivity sweep over *ProportionBiogas* (0.33–1.00) and butyric acid price (USD 2.00–5.00/kg) confirms this: the response surface is strongly unimodal at *ProportionBiogas* = 1, with no saddle point or local maximum at interior allocations. The profit function is monotonically increasing in the biogas proportion across the entire feasible allocation space under current market parameters.

4.4 Sensitivity Analysis

A one-at-a-time sensitivity analysis varied financial parameters ($\pm 5\%$ to $\pm 100\%$) and technical conversion factors ($\pm 10\%$ and $\pm 90\%$), reporting the Mean Net Surplus over the post-warmup period 2008–2030 in MMUSD. Results are summarized in Tables 6 and 7. To complement the tabular results, Figures 4, 5, and 6 present cross-scenario comparisons of the net profit trajectories under $\pm 50\%$ variation in price, conversion factor, and processing cost respectively, providing a direct visual comparison of how each pathway responds to moderate structural shocks.

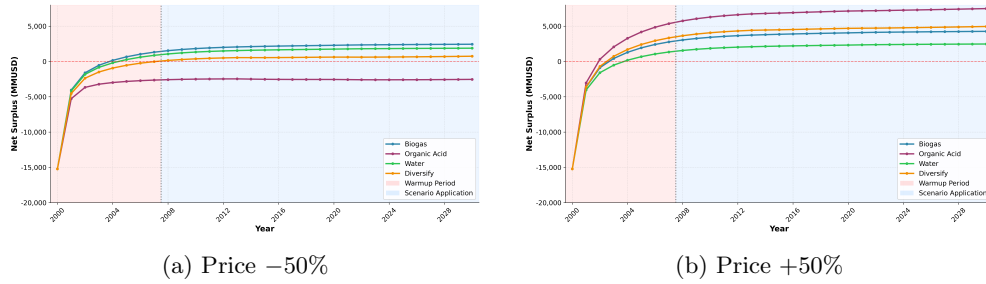


Fig. 4: Cross-scenario comparison of net profit trajectories under $\pm 50\%$ price variation for all four valorization pathways over the 30-year horizon. All monetary values in million USD (MMUSD).

Across pathways, a consistent risk hierarchy emerges that reinforces the comparative profitability ranking established in Section 4.2. The cross-scenario comparisons at $\pm 50\%$ (Figures 4–6) reveal that the profitability ranking is not invariant to structural shocks: under price -50% , butyric acid enters persistent negative territory (mean net surplus $-2,543.7$ MMUSD over the post-warmup period), while biogas retains $2,191.2$ MMUSD, water recovery retains $1,661.3$ MMUSD, and diversification retains only 563.4 MMUSD. Under price $+50\%$, the ranking partially reverses for butyric acid ($6,955.9$ MMUSD), which surpasses diversification ($4,555.4$ MMUSD) and approaches biogas ($3,919.4$ MMUSD), illustrating that butyric acid’s high sensitivity cuts in both directions.

Biogas exhibits the highest absolute surplus and the strongest resilience across all three parameter categories (Table 6). Price variation of $\pm 25\%$ produces a mean net surplus range of $2,623.3$ to $3,487.3$ MMUSD (variation of 864.1 MMUSD), and even under complete price collapse (-100%) the pathway retains $1,327.2$ MMUSD,

Table 6: Sensitivity Analysis: Mean Net Surplus (2008–2030) under Parameter Variation (Part 1 – Biogas & Organic Acid)

Scenario Parameter	Base Value	Range Tested			Mean Net Surplus 2008–2030 (MMUSD)	Var. (MMUSD)
Biogas						
Biogas price	\$0.50/Nm ³	\$0.00–1.00 (+100%)	(–100%	to	1,327.2 to 4,783.5	3,456.3
		\$0.25–0.75 (+50%)	(–50%	to	2,191.2 to 3,919.4	1,728.1
		\$0.38–0.63 (+25%)	(–25%	to	2,623.3 to 3,487.3	864.
		\$0.48–0.53 (–5% to +5%)			2,968.9 to 3,141.7	172.8
Biogas conversion factor	12.115 Nm ³ /m ³	1.21–23.02 (+90%)	(–90%	to	1,586.4 to 4,524.2	2,937.9
		6.06–18.17 (+50%)	(–50%	to	2,320.8 to 3,789.8	1,468.9
		9.09–15.14 (+25%)	(–25%	to	2,688.1 to 3,422.5	734.5
		10.90–13.33 (+10%)	(–10%	to	2,981.9 to 3,128.8	146.9
Processing cost	\$0.074/Nm ³	\$0.00–0.15 (+100%)	(–100%	to	2,796.1 to 3,314.5	518.4
		\$0.04–0.11 (+50%)	(–50%	to	2,925.7 to 3,184.9	259.2
		\$0.06–0.09 (+25%)	(–25%	to	2,990.5 to 3,120.1	129.6
		\$0.07–0.08 (–5% to +5%)			3,042.4 to 3,068.3	25.9
Organic Acid						
Product price	\$2.89/kg	\$0.00–5.78 (+100%)	(–100%	to	–7,293.5 to 11,705.6	18,999.1
		\$1.45–4.34 (+50%)	(–50%	to	–2,543.7 to 6,955.9	9,499.6
		\$2.17–3.61 (+25%)	(–25%	to	–168.8 to 4,580.9	4,749.7
		\$2.74–3.04 (–5% to +5%)			1,731.0 to 2,681.1	950.0
Conversion factor	0.0106 kg/m ³	0.0011–0.0201 (+90%)	(–90%	to	1,746.0 to 2,666.3	920.3
		0.0053–0.0159 (+50%)	(–50%	to	1,976.1 to 2,436.2	460.1
		0.0080–0.0133 (+25%)	(–25%	to	2,091.0 to 2,321.0	230.0
		0.0095–0.0117 (+10%)	(–10%	to	2,183.0 to 2,229.3	46.3
Processing cost	\$2.75/kg	\$0.00–5.50 (+100%)	(–100%	to	–6,833.3 to 11,245.4	18,078.7
		\$1.38–4.13 (+50%)	(–50%	to	–2,313.7 to 6,725.7	9,039.4
		\$2.06–3.44 (+25%)	(–25%	to	–53.9 to 4,465.8	4,519.7
		\$2.61–2.89 (–5% to +5%)			1,754.2 to 2,658.0	903.8

driven by the structural cost advantage of UASB infrastructure. Conversion factor variation of $\pm 25\%$ produces a 734.5 MMUSD swing (2,688.1 to 3,422.5 MMUSD), remaining fully within profitable territory across the entire tested range including the extreme $\pm 90\%$ bounds (1,586.4 to 4,524.2 MMUSD). Processing cost sensitivity is the narrowest of all categories: at $\pm 25\%$ the variation is only 129.6 MMUSD (2,990.5 to 3,120.1 MMUSD), confirming that biogas profitability is governed by the revenue side rather than operational costs (Figure 5).

Table 7: Sensitivity Analysis: Mean Net Surplus (2008–2030) under Parameter Variation (Part 2 – Water Recovery & Diversification)

Scenario Parameter	Base Value	Range Tested	Mean Net Surplus 2008–2030 (MMUSD)	Var. (MMUSD)
Water Recovery				
Water value	\$2.81/m ³	\$0.00–5.62 (–100% to +100%)	1,380.7 to 2,503.0	1,122.3
		\$1.41–4.22 (–50% to +50%)	1,661.3 to 2,222.4	561.2
		\$2.11–3.51 (–25% to +25%)	1,801.6 to 2,082.1	280.6
		\$2.67–2.95 (–5% to +5%)	1,913.8 to 1,969.9	56.1
Recovery factor	0.70	0.07–1.33 (–90% to +90%)	1,586.4 to 2,297.3	710.9
		0.35–1.05 (–50% to +50%)	1,764.1 to 2,119.6	355.5
		0.53–0.88 (–25% to +25%)	1,853.0 to 2,030.7	177.7
		0.63–0.77 (–10% to +10%)	1,924.1 to 1,959.6	35.5
Treatment cost	\$1.03/m ³	\$0.00–2.06 (–100% to +100%)	1,736.2 to 2,147.6	411.4
		\$0.52–1.55 (–50% to +50%)	1,839.0 to 2,044.7	205.7
		\$0.77–1.29 (–25% to +25%)	1,890.4 to 1,993.3	102.8
		\$0.98–1.08 (–5% to +5%)	1,931.6 to 1,952.1	20.6
Diversification (Balanced Equal Weighting)				
Combined prices	Weighted index	–100% to +100%	–1,432.6 to 6,551.5	7,984.1
		–50% to +50%	563.4 to 4,555.4	3,992.0
		–25% to +25%	1,561.5 to 3,557.5	1,996.0
		–5% to +5%	2,359.9 to 2,759.1	399.2
Conversion factor	Mixed yield	–90% to +90%	1,746.0 to 3,373.3	1,627.3
		–50% to +50%	2,152.7 to 2,966.4	813.7
		–25% to +25%	2,356.2 to 2,763.0	406.8
		–10% to +10%	2,518.9 to 2,600.3	81.4
Processing cost	Weighted index	–100% to +100%	–618.8 to 5,738.0	6,356.7
		–50% to +50%	970.4 to 4,148.7	3,178.3
		–25% to +25%	1,765.0 to 3,354.2	1,589.2
		–5% to +5%	2,400.7 to 2,718.4	317.7

Water recovery is the most stable pathway in absolute terms across all three parameter categories (Table 7). Price variation of $\pm 25\%$ produces only a 280.6 MMUSD swing (1,801.6 to 2,082.1 MMUSD), and even at $\pm 90\%$ the recovery factor variation spans just 710.9 MMUSD (1,586.4 to 2,297.3 MMUSD) — comparable in magnitude to biogas factor sensitivity but at a lower absolute level. Treatment cost sensitivity across the full $\pm 100\%$ range spans 411.4 MMUSD (1,736.2 to 2,147.6 MMUSD), the narrowest cost range of all pathways. Even under complete elimination of water value (-100%), the pathway retains a mean surplus of 1,380.7 MMUSD, confirming water recovery as a robust defensive complement to biogas whose financial case does not depend on favorable technical or market conditions (Figure 4).

Butyric acid exhibits the highest sensitivity of all pathways across all three parameter categories (Table 6). Price variation of $\pm 25\%$ generates a 4,749.7 MMUSD swing (-168.8 to 4,580.9 MMUSD), and at -50% the mean net surplus turns negative ($-2,543.7$ MMUSD). Processing cost sensitivity mirrors this pattern: at $\pm 50\%$ the range spans 9,039.4 MMUSD ($-2,313.7$ to 6,725.7 MMUSD), and at -100% cost rises

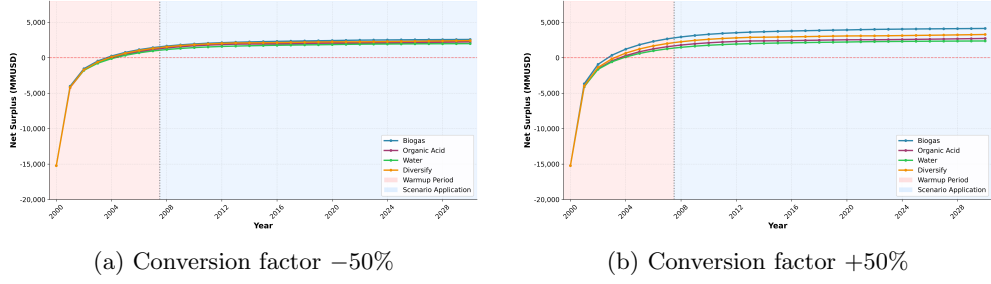


Fig. 5: Cross-scenario comparison of net profit trajectories under $\pm 50\%$ conversion factor variation for all four valorization pathways over the 30-year horizon. All monetary values in million USD (MMUSD).

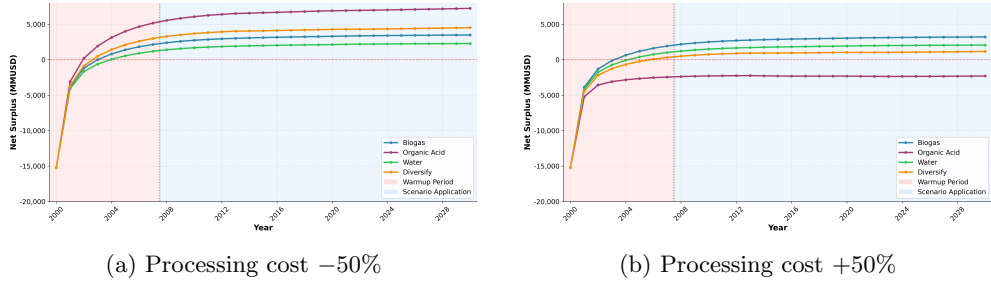


Fig. 6: Cross-scenario comparison of net profit trajectories under $\pm 50\%$ processing cost variation for all four valorization pathways over the 30-year horizon. All monetary values in million USD (MMUSD).

to 11,245.4 MMUSD while at $+100\%$ it falls to $-6,833.3$ MMUSD. In contrast, conversion factor sensitivity is comparatively contained: at $\pm 50\%$ the variation is only 460.1 MMUSD (1,976.1 to 2,436.2 MMUSD), and even at the extreme $\pm 90\%$ bounds the range is 920.3 MMUSD (1,746.0 to 2,666.3 MMUSD). This asymmetry confirms that the primary risk driver for butyric acid is market price and processing cost volatility rather than technical yield uncertainty. Deployment should therefore be pursued selectively by operators with established offtake agreements; market aggregation mechanisms or long-term supply contracts are prerequisites for sector-wide adoption.

The diversification scenario shows intermediate sensitivity across all three categories (Table 7). Combined price variation of $\pm 25\%$ produces a 1,996.0 MMUSD swing (1,561.5 to 3,557.5 MMUSD), and the pathway enters negative surplus territory only under the most extreme price shock (-100% , yielding $-1,432.6$ MMUSD). Conversion factor sensitivity at $\pm 50\%$ spans 813.7 MMUSD (2,152.7 to 2,966.4 MMUSD), remaining fully positive across all tested bounds. Processing cost sensitivity at $\pm 50\%$ spans 3,178.3 MMUSD (970.4 to 4,148.7 MMUSD) — substantially narrower than butyric acid’s 9,039.4 MMUSD at the same level, confirming that equal-weight diversification provides meaningful downside protection by diluting butyric acid’s extreme

cost sensitivity with the more stable biogas and water recovery pathways. Together, Figures 4–6 and Tables 6–7 confirm that the risk-return trade-off favors biogas as the primary pathway and water recovery as the defensive complement, with diversification offering a rational alternative only for operators who prioritize downside protection over maximum expected surplus.

4.5 Policy and Strategic Implications

The results support a tiered incentive architecture rather than the equal-weight diversification that characterizes most current policy frameworks. Biogas is the most appropriate entry point for vinasse valorization across mill sizes: its wide price-cost spread, strongest resilience to price shocks, and direct contribution to SDG 7 by displacing fossil fuels make it the highest-priority target for initial infrastructure support and RenovaBio credit sequencing (Sica et al., 2023). Water recovery merits attention as a defensive complement whose financial case does not depend on process efficiency improvements but on the regulatory trajectory of water pricing: as freshwater scarcity intensifies in Brazilian sugarcane-growing regions, abstraction charges and water trading mechanisms will strengthen the financial case independently of any technological advance, making it well-suited for blended finance or green bond instruments aligned with SDG 6. Butyric acid deployment should be contingent on market maturation; sector-wide adoption without established offtake agreements could depress prices toward the -50% threshold at which profitability turns negative, triggering the market saturation dynamics that the single-facility model boundary does not capture.

The biogas dominance result is conditional on current market prices and regulatory stability. Staged infrastructure capable of accommodating both UASB reactors and evaporative concentration units would preserve optionality if CBIO credit trajectories or water abstraction charges shift in favor of water recovery. Formal optimization challenging the policy tendency toward equal-weight diversification supports initial support for biogas and water recovery to establish early financial stability, followed by transitional support for butyric acid as chemical markets develop, positioning the sector as a concrete contributor to both SDG 6 and SDG 7 while avoiding the market saturation effects that simultaneous adoption of all pathways could trigger.

4.6 Model Limitations and Future Research Directions

The single-facility boundary treats by-product prices as exogenous, omitting feedback between cumulative sector-wide supply and market prices. This is particularly consequential for butyric acid, where sector-wide adoption could depress prices beyond the -50% threshold at which profitability turns negative. Future extensions should incorporate multi-facility supply-demand equilibrium structures with price-response functions calibrated from demand elasticity data (IRENA, 2017; Peiter et al., 2019; Suazo et al., 2023). The regulatory environment is also abstracted: RenovaBio CBIO credit prices, feed-in tariffs, and water abstraction charges are dynamic drivers each capable of displacing the optimal allocation; future work should incorporate these as stochastic variables with transition probabilities derived from Brazil’s legislative history. Further extensions include multi-objective optimization balancing financial

return against water security and carbon reduction indicators, and disaggregation to mill-level or spatially explicit configurations to capture regional strategies and inter-mill competition dynamics.

5 Conclusion

This study developed and validated an integrated system dynamics model to evaluate circular economy strategies for vinasse valorization within the Brazilian sugarcane ethanol sector, addressing the absence of frameworks that dynamically couple byproduct valorization economics to the production system over multidecade horizons (Section 3). Structural validity was established prior to behavior validation through systematic unit checking, parameter verification, and three extreme condition tests. The model was then validated against 23 years of Brazilian historical data (2000–2023), achieving in-sample MAPEs of 7.98% for ethanol production and 4.82% for crop production, and out-of-sample MAPEs of 8.48% and 6.30% respectively, all below the 10% high-accuracy threshold (Appendix B). Five scenarios were simulated over a 30-year horizon, yielding four principal findings.

All circular economy pathways are economically viable (Section 4.2). All pathways enter the post-warmup period at Year 8 already operating with positive margins, with long-term margins stabilizing between 8.91% and 17.76% and cumulative profits over the 30-year horizon ranging from 22,395 MMUSD (diversification) to 61,638 MMUSD (biogas). Early losses during the initialization period should not constitute a barrier to adoption when evaluated over appropriate planning horizons.

Among individual pathways, biogas emerges as the superior strategy, combining the highest long-term profitability (17.76% at Year 30, cumulative profit 61,638 MMUSD) with the strongest resilience to price shocks, retaining a mean net surplus of 2,191.2 MMUSD even under a -50% price shock and remaining profitable across the full tested parameter range (Section 4.4). As a renewable energy source displacing fossil fuels for steam generation, biogas directly advances SDG 7 on Affordable and Clean Energy, reinforcing its status as the most strategically and sustainably compelling entry point for circular biorefinery adoption. Water recovery stabilizes at 12.34% margin with a cumulative profit of 31,891 MMUSD, providing the most stable returns across all tested conditions, with a mean net surplus of 1,380.7 MMUSD even under complete elimination of water value. Its simultaneous contribution to revenue generation and irrigation cost reduction, combined with its alignment with SDG 6 on Clean Water and Sanitation, strengthens its long-term financial case as freshwater scarcity intensifies in Brazilian sugarcane-growing regions (Section 4.2). Butyric acid, while commercially viable at the mill scale (8.91% margin), exhibits extreme price and cost sensitivity — the full $\pm 100\%$ price range spans 13,191.9 MMUSD — and requires market-structuring interventions such as offtake agreements or price floors before sector-wide adoption.

Equal-weight diversification does not outperform targeted biogas investment, stabilizing at 10.57% net profit and 22,395 MMUSD cumulative profit, 7.2 pp below the biogas ceiling, because resource allocation trade-offs dilute the highest-return pathway without a commensurate risk reduction (Section 4.2). Formal optimization via

OptQuest confirms this: the solver converged over 500 iterations to a corner solution allocating all vinasse to biogas, with a cumulative profit of 61,638 MMUSD — 175% higher than the diversification scenario — demonstrating that no mixed allocation outperforms full biogas dedication under current market parameters (Section 4.3). This conclusion is conditional on profit maximization under Brazilian market prices and may shift where non-financial criteria such as water security obligations or underdeveloped biogas infrastructure alter pathway rankings.

These results support a tiered incentive architecture prioritizing biogas and water recovery to establish early financial stability, followed by transitional support for butyric acid as chemical markets develop, while avoiding the market saturation effects that simultaneous adoption of all pathways could trigger (Section 4.5). The alignment of water recovery with SDG 6 further suggests that public finance instruments such as blended finance or green bonds could support infrastructure investments whose social value exceeds current market prices. Priority future extensions include endogenizing by-product pricing, incorporating stochastic policy variables under RenovaBio, developing multi-objective optimization formulations, and disaggregating the model to mill-level or spatially explicit configurations (Section 4.6).

Together, a structurally validated feedback-rich SDM, a 23-year empirical foundation, comparative scenario evidence revealing non-obvious diversification dynamics, and a formally optimized allocation provide stakeholders with an evidence-based decision support framework confirming that circular economy approaches can substantially improve the economic and environmental sustainability of the sugarcane-ethanol sector while contributing to measurable progress toward both SDG 6 on Clean Water and Sanitation and SDG 7 on Affordable and Clean Energy.

Appendix A Mathematical Formulation of the System Dynamics Model

This appendix presents the complete stock-and-flow formulation of each module in the system dynamics model, complementing the conceptual description in Section 3. All equations are integrated continuously over the simulation horizon $t \in [0, 30]$ years using a Euler method. Explicit units are provided for all variables to document dimensional consistency.

A.1 Consumer Market Module

Population is obtained by linear interpolation from a lookup table containing historical Brazilian census data and demographic projections for the period 2000–2030 (Equation A1). This formulation replaces the exponential birth-death structure with an empirically grounded trajectory that captures the observed deceleration in Brazilian population growth without imposing a structural form that diverges from historical data. Historical values for the period 2000–2025 were obtained from Expansión/Datosmacro (Expansión/Datosmacro (2024)). Projected values for 2026–2030 are drawn from IMF estimates published via Statista (IMF (2025)), last accessed March 2026.

$$population(t) = \text{LOOKUP}(t), \quad t \in [0, 30] \quad (\text{A1})$$

where *population* is expressed in persons and *t* represents years elapsed since 2000. Table A1 reports the complete set of values used in the lookup function.

Table A1: Brazilian population lookup table used in the Consumer Market Module. Historical values (2000–2025) from Expansión/Datosmacro ([Expansión/Datosmacro \(2024\)](#)); projected values (2026–2030) from IMF estimates via Statista ([IMF \(2025\)](#)).

Year	<i>t</i>	Population	Source
2000	0	174,018,282	Historical
2001	1	176,301,203	Historical
2002	2	178,503,484	Historical
2003	3	180,622,688	Historical
2004	4	182,675,143	Historical
2005	5	184,688,101	Historical
2006	6	186,653,106	Historical
2007	7	188,552,320	Historical
2008	8	190,367,302	Historical
2009	9	192,079,951	Historical
2010	10	193,701,929	Historical
2011	11	195,284,734	Historical
2012	12	196,876,111	Historical
2013	13	198,478,299	Historical
2014	14	200,085,127	Historical
2015	15	201,675,532	Historical
2016	16	203,218,114	Historical
2017	17	204,703,445	Historical
2018	18	206,107,261	Historical
2019	19	207,455,459	Historical
2020	20	208,660,842	Historical
2021	21	209,550,294	Historical
2022	22	210,306,415	Historical
2023	23	211,140,729	Historical
2024	24	211,998,573	Historical
2025	25	212,812,405	Historical
2026	26	213,600,000	Projected
2027	27	214,400,000	Projected
2028	28	215,200,000	Projected
2029	29	216,000,000	Projected
2030	30	216,800,000	Projected

The demand volume is calculated from population, the average ethanol usage rate per person, and a volume conversion factor (Equation A2):

$$DemandVolume = \frac{EthanolUsage}{volConversionFactor} \times population \quad [\text{thousand m}^3/\text{year}] \quad (\text{A2})$$

where *DemandVolume* is expressed in thousand m³/year, *EthanolUsage* in thousand m³/person, and *volConversionFactor* is a dimensionless scaling factor of 1,000. The *EthanolUsage* parameter was estimated as the ratio of national ethanol production to total population over the period 2000–2023 ([UNICAdata \(2024\)](#)).

A.2 Ethanol Production Module — Ethanol

The supply-side SFD (Figure A1) integrates demand volume, production, and inventory management. The Ethanol Inventory stock changes via inflows from domestic production and imports triggered by shortage signals, and via outflows from sales (Equation A3). Production is proportional to crop availability through a yield coefficient (Equation A4). Sales are constrained by available inventory and demand volume, where both terms are expressed in thousand m³ ensuring dimensional consistency (Equation A5). Shortfalls trigger imports (Equations A6–A7).

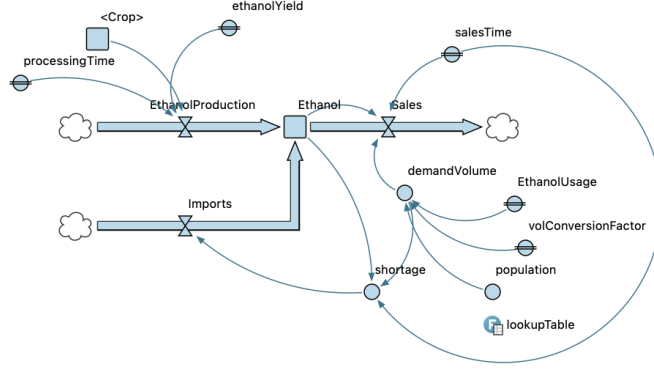


Fig. A1: Stock-and-Flow Diagram for Ethanol Inventory and Supply.

$$\frac{d(Ethanol)}{dt} = EthanolProduction + Imports - Sales \quad [\text{thousand m}^3/\text{year}] \quad (A3)$$

$$EthanolProduction = \frac{Crop}{ProcessingTime} \times ethanolYield \quad [\text{thousand m}^3/\text{year}] \quad (A4)$$

$$Sales = \min\left(\frac{Ethanol}{SalesTime}, DemandVolume\right) \quad [\text{thousand m}^3/\text{year}] \quad (A5)$$

$$shortage = \max\left(0, DemandVolume - \frac{Ethanol}{SalesTime}\right) \quad [\text{thousand m}^3/\text{year}] \quad (A6)$$

$$Imports = shortage \quad [\text{thousand m}^3/\text{year}] \quad (A7)$$

Both *Ethanol* (stock) and *DemandVolume* (auxiliary variable) are expressed in thousand m³, ensuring that the *Sales* and *shortage* equations are dimensionally consistent.

A.3 Ethanol Production Module — Crops

The crop production SFD (Figure A2) defines two primary stocks: *YoungCrop* (biomass in growth phase, thousand tons) and *Crop* (harvest-ready biomass, thousand tons), governed by Equations A9–A10.

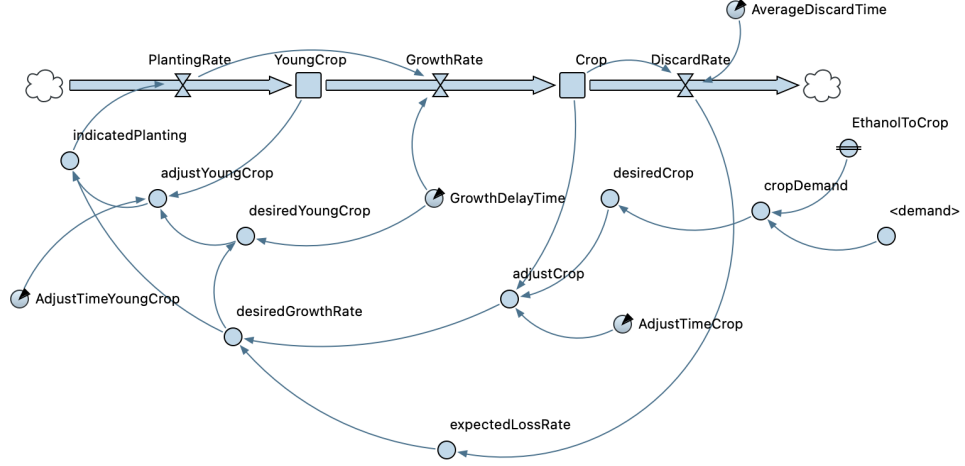


Fig. A2: Stock-and-Flow Diagram for Crop Production.

Crop maturation is represented using a third-order material delay (Equation A8), implemented via the `delay3` operator (Sterman (2000)). This formulation captures the biological reality that sugarcane does not mature instantaneously: planted biomass passes through three sequential internal stages before becoming harvest-ready, producing an S-shaped response in the output flow that is more realistic than a simple first-order delay. The mean transit time through all three stages is governed by *GrowthDelayTime*, and the output of the delay — the rate at which young crop matures into harvest-ready crop — is denoted *GrowthRate*. The stocks *YoungCrop* and *Crop* evolve according to the inflow and outflow of this maturation process:

$$GrowthRate = \text{delay3}(PlantingRate, GrowthDelayTime) \quad [\text{thousand tons/year}] \quad (\text{A8})$$

$$\frac{d(YoungCrop)}{dt} = PlantingRate - GrowthRate \quad [\text{thousand tons/year}] \quad (\text{A9})$$

$$\frac{d(Crop)}{dt} = GrowthRate - DiscardRate \quad [\text{thousand tons/year}] \quad (\text{A10})$$

$$DiscardRate = \frac{Crop}{AverageDiscardTime} \quad [\text{thousand tons/year}] \quad (\text{A11})$$

Target stock levels are set by desired production goals, and feedback control mechanisms adjust planting decisions to close gaps between actual and desired stocks following the stock-and-flow adjustment pattern (Sterman (2000)) (Equations A12–A20):

$$cropDemand = DemandVolume \times EthanolToCrop \quad [\text{thousand tons/year}] \quad (\text{A12})$$

$$desiredCrop = cropDemand \quad [\text{thousand tons/year}] \quad (\text{A13})$$

$$adjustCrop = \frac{desiredCrop - Crop}{AdjustTimeCrop} \quad [\text{thousand tons/year}] \quad (A14)$$

$$ExpectedLossRate = DiscardRate \quad [\text{thousand tons/year}] \quad (A15)$$

$$DesiredGrowthRate = \max(0, ExpectedLossRate + adjustCrop) \quad [\text{thousand tons/year}] \quad (A16)$$

$$desiredYoungCrop = DesiredGrowthRate \times GrowthDelayTime \quad [\text{thousand tons}] \quad (A17)$$

$$adjustYoungCrop = \frac{desiredYoungCrop - YoungCrop}{AdjustTimeYoungCrop} \quad [\text{thousand tons/year}] \quad (A18)$$

$$indicatedPlanting = adjustYoungCrop + DesiredGrowthRate \quad [\text{thousand tons/year}] \quad (A19)$$

$$PlantingRate = \max(0, indicatedPlanting) \quad [\text{thousand tons/year}] \quad (A20)$$

The *indicatedPlanting* equation (Equation A19) follows the stock-and-flow adjustment pattern (Sterman (2000)), combining a gap-closing signal (*adjustYoungCrop*) that drives the young crop stock toward its desired level, and a flow replacement signal (*DesiredGrowthRate*) that maintains the throughput required to sustain the mature crop stock. Both terms are expressed in thousand tons/year, ensuring dimensional consistency. The *ExpectedLossRate* is set equal to the current *DiscardRate* (Equation A15), representing a first-order approximation of anticipated crop losses used to adjust planting targets.

A.4 By-product Production Module

The BPM (Figure A3) transforms vinasse into value-added byproducts, operationalizing the circular economy logic of the biorefinery. Vinasse generation is computed as an auxiliary variable proportional to ethanol output in each period (Equation A21), consistent with the continuous-flow processing assumption of industrial distillery operations where vinasse is treated as generated rather than stored in bulk. The vinasse generated in each period is allocated across three parallel conversion pathways through proportion parameters, each governed by a conversion efficiency factor (Equations A22–A24).

$$VinasseProduction = EthanolProduction \times VinasseYield \quad [\text{thousand m}^3/\text{year}] \quad (A21)$$

$$BiogasProduction = VinasseProduction \times ProportionBiogas \times BiogasFactor \quad [\text{thousand Nm}^3/\text{year}] \quad (A22)$$

$$OrganicAcidProduction = VinasseProduction \times ProportionOrganicAcid \times OrganicAcidFactor \quad [\text{thousand kg/year}] \quad (A23)$$

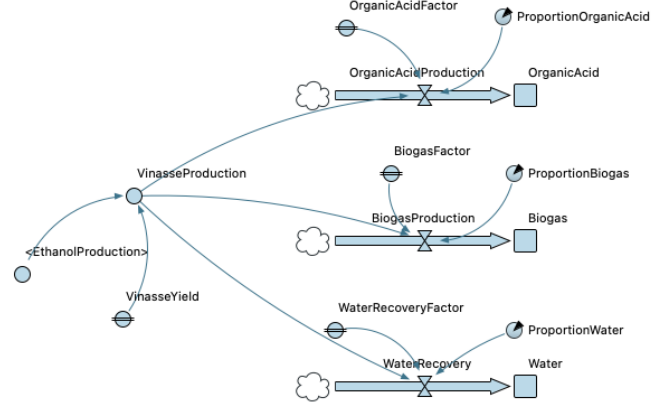


Fig. A3: Stock-and-Flow Diagram for By-product Production.

$$WaterRecovery = VinasseProduction \times ProportionWater \times WaterRecoveryFactor \quad [\text{thousand m}^3/\text{year}] \quad (A24)$$

VinasseProduction is an auxiliary variable computed from the ethanol production flow in each integration step; it is not a stock accumulator. By-product prices for biogas, butyric acid, and recovered water are treated as exogenous parameters, consistent with the single-facility model boundary in which the output of an individual mill does not influence prevailing market prices. The causal coupling between modules is unidirectional: ethanol production determines vinasse generation, which drives byproduct revenues, but byproduct revenues do not feed back into upstream agricultural or ethanol production decisions.

Appendix B Parameter Calibration Procedure

B.1 Model Verification

Model verification involved systematic unit checking across all equations, assigning explicit units to every variable and flow to identify and correct dimensional mismatches. Key corrections included renaming the demand variable to *DemandVolume* (thousand m³) to ensure consistency in the inventory comparison equations, simplifying *ExpectedLossRate* to equal *DiscardRate* directly, and documenting the delay3 operator as an Erlang-3 pipeline decomposed into three internal levels. All conversion factors were verified for dimensional integrity prior to empirical testing.

B.2 Parameter Calibration

Four parameters were estimated using the OptQuest solver built into AnyLogic, configured to minimize the root mean square error (RMSE) between simulated and historical

crop production values over 500 fixed iterations (Equation B25). The RMSE was used as the calibration objective to penalize large deviations during the calibration window; the MAPE metric (Equation B26) is then used separately to assess validation accuracy, as it provides a scale-independent measure suitable for comparing performance across variables with different magnitudes.

$$\min_{\theta} \sqrt{\frac{1}{11} \sum_{t=8}^{18} \left(Crop_{sim}(t; \theta) - cropHistoric(t) \right)^2} \quad (B25)$$

where $t = 8$ corresponds to year 2008 and $t = 18$ to year 2018, covering 11 observations within the calibration window. The first eight years ($t = 0$ to $t = 7$, corresponding to 2000–2007) were used as a warm-up period to initialize all stocks before calibration begins.

Search bounds were defined from literature-derived ranges: *AdjustTimeYoungCrop* and *GrowthDelayTime* bounded between 1 and 2 (Oré et al. (2022)), *AdjustTimeCrop* between 3 and 4 (Kittilertpaisan and Pathumnakul (2017)), and *AverageDiscardTime* between 5 and 8 (Xu, Wang, Lu, Zeng, and Que (2021)). This literature-grounded strategy ensured the solver explored only agronomically plausible regions of the parameter space. The calibrated values are reported in Table 4. The exclusive use of crop production as the calibration target reflects that the four estimated parameters govern only the agricultural sub-system; ethanol production dynamics are driven by the fixed *ethanolYield* coefficient, and population dynamics are represented by a lookup table sourced from Expansión/Datosmacro (Expansión/Datosmacro (2024)) for 2000–2025 and from IMF estimates via Statista (IMF (2025)) for 2026–2030, neither of which requires data-driven estimation.

B.3 Validation Metric

Accuracy was assessed using the Mean Absolute Percentage Error (MAPE) (Akha-van and Gonçalves (2021); De Myttenaere et al. (2016); Yuniaristanto et al. (2024)), defined as:

$$MAPE = \frac{1}{n} \sum_{t=1}^n \left| \frac{A_t - F_t}{A_t} \right| \times 100 \quad (B26)$$

where n is the number of observations, A_t is the actual value, and F_t is the simulated value at time t . In the context of SDM validation, MAPE below 10% is considered highly accurate, 10% to 20% indicates acceptable accuracy for complex socio-technical systems, and above 20% requires recalibration (Stermann (2000)).

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